Know When to Fold: Risk-Controlled Abstention for Protein–Ligand Co-Folding under Novelty Shift

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Abstract

Protein–ligand co-folding models (AlphaFold3, Boltz, Chai) report confidence scores that correlate with pose accuracy, but correlation is not a decision rule, and it is weakest on the novel pockets and chemotypes drug discovery cares about. We build `foldgate`, a training-free layer that turns frozen co-folding confidence into a risk-controlled accept/abstain gate, and audit where its guarantee fails. Calibrating on familiar targets and deploying on novel ones drives realized selective error to 0.55 against a 0.20 target (AF3; 0.48 to 0.66 across five models). A per-stratum feasibility analysis shows this is not a tuning problem: of 50 model-by-stratum cells, 35 admit *no* threshold on the frozen score that reaches the target at any coverage, robust to an exact binomial lower bound, so abstention is the only certified action there. An exact decomposition of the realized risk explains why and yields a measurable drift diagnostic that selects the repair from a labeled probe. Group-conditional recalibration restores per-stratum control, retaining 73% of AF3 poses at certified risk ≤ 0.20 under leakage-free target-grouped cross-validation where the native gate retains 20%, at a price we measure: a median of 38 in-stratum labels on the familiar strata, and, on the novel tail, no repair by stratification alone. A public deployment-computable proxy for the stratifier tracks the oracle on low and moderate novelty but loosens control on the novel tail, so for closed-training models the repair is deployable one stratum out and the extreme tail remains uncertifiable. The layer runs on released predictions on CPU and ships as certificate cards per model and novelty stratum.

1 Introduction

Co-folding models predict a protein–ligand complex and emit confidence scores (interface predicted TM-score, ipTM; ranking score; predicted aligned error). A practitioner who wants to act on a predicted pose needs a decision rule that carries an error guarantee. Conformal selective prediction supplies one: accept a pose when its score clears a threshold, abstain otherwise, and certify that the error rate among accepted poses stays below a target α with high probability [7, 21]. The catch is that this guarantee assumes exchangeability, that a new target is statistically interchangeable with the calibration examples, and that breaks precisely where deployment matters, on ligands and pockets dissimilar to the training set. In drug-discovery terms the model is most overconfident on the never-before-seen molecules a program cares about, so the operational question is not average accuracy but whether a single delivered pose can be trusted.

We make three contributions, all training-free with respect to the co-folding model and all computed from released predictions on CPU. **(1)** We give a finite-sample selective-risk gate over frozen co-folding confidence, show its guarantee collapses under novelty shift, and map the collapse with a per-stratum *feasibility frontier* that decays to zero on novel strata; reliability drift attributes the collapse to concept shift (Sec. 3–4). **(2)** We prove an exact decomposition of the realized selective risk into a covariate-corrected reference and an accept-region concept gap that no training-free reweighting can move, turning it into a measurable diagnostic that selects, from a labeled probe, which repair a deployment region admits (Sec. 5). **(3)** We repair what is repairable with group-conditional calibration and a label-free worst-subpopulation certificate, match the certifier to the loss, and *price* the repair: the label cost as a design curve, a deployment-computable proxy stratifier, and the honest bottom line that the extreme novel tail is uncertifiable by stratification alone (Sec. 6). A leakage-free utility evaluation and a decision-curve analysis follow (Sec. 7).

The closest prior work is CoDrug [14], which conformalizes a *scalar* molecular property under covariate shift with the same weighted-conformal tool [3] our decomposition shows concept shift defeats in this

Table 1: Notation.

L	loss $\mathbf{1}[\text{RMSD} > 2\,\text{Å}]$	α, δ	risk target, failure prob. $(0.20, 0.10)$
s, τ	score, accept threshold ($s \geq \tau$)	c	coverage (accepted fraction)
ν, S_ν	novelty axis, stratum	η_P, η_Q	source/target error conditional $P(L{=}1 \mid X)$
R_{ref}	covariate-corrected reference risk	$\bar{\Delta}_c$	accept-region concept gap
$D(\nu)$	reliability drift at stratum ν	$m^\star, \beta^\star$	certified min. subpop. mass, CVaR level
$\rho^\star$	DRO robustness radius	n_g	in-stratum labels

domain; we are not aware of a prior conformal treatment of co-folding pose correctness, or of training-set similarity used as the shift variable for it. Confidence-tracks-accuracy studies for co-folding virtual screening [10] rank by ipTM but supply neither a coverage guarantee nor an abstention rule, which is the gap this layer fills. A reweighting-invariant selective-risk floor of the same abstract character was shown independently and concurrently for LLM abstention [19] (arXiv preprint, June 2026); ours is its co-folding specialization, localized per novelty stratum with a matching in-stratum achievability bracket. Conformal selection [15, 16] (the latter an arXiv preprint) is complementary: it certifies which *compounds* to advance given a labeled calibration set, while this layer certifies the *pose* correctness a co-folding screen rests on, training-free.

2 Data and definitions

Unit and loss. The decision-relevant unit is a delivered top-1 pose per system with a native confidence s. The loss is $L = \mathbf{1}[\text{symmetry-corrected ligand-RMSD} > 2\,\text{Å}]$ against the released BiSyRMSD label, so a certified selective risk is a bound on the accepted-set error rate. We use L throughout (Table 1). RMSD is the shipped symmetry-corrected BiSyRMSD (receptor superposition, ligand-in-place); recomputed geometry is used only where noted (Table 2).

Novelty axes. We stratify by training-set similarity along two structural axes: a ligand axis (ECFP4 Morgan Tanimoto to the nearest training ligand) and a pocket axis (SuCOS shape $\times$ pocket query-coverage), both released by Runs N' Poses. Each axis is binned into quartiles S_0 (familiar) to S_3, with systems that have no training analog (missing similarity, 30–34% of poses) forming a fifth no-analog stratum S_4. The quartile edges are fixed a priori from the similarity marginal; Table S1 lists n and the similarity range each bin spans, per model and axis. We treat structural similarity, not deposition recency, as the operative shift variable, for a reason the data forces (Sec. 3).

Cohort. We evaluate five models (AF3 [11], Boltz-1, Boltz-1x [20], Chai-1 [12], Protenix [13]) on Runs N' Poses [8]. Figure 1 is the reduction chain: 13,535 raw delivered poses (six models) minus the 933 Boltz-2 poses used only as an ungoverned affinity comparator (Sec. 7) give 12,602 governed poses over 2,425 systems; deduplicating to one pose per (system, model) gives 11,254 independent target-labels for the five governed models. Certificates that must be per deployed model use the target-label unit, since several ligand instances of one system fail together and pooling them as independent draws is anti-conservative. The cohort is chemically diverse and not a single-family monoculture: 55% drug-like (300–500 Da) with a 24% fragment tail, 2,412 distinct receptors, and no PDB header class above 21%.

Deployment cost. The native-score gate is CPU-only and consumes one released prediction per target. An optional combined score (a calibration-only gradient-boosted map from cheap per-prediction features to $P(\text{correct})$: confidence, interface ipTM, ensemble spread, cross-model agreement, PoseBusters validity, ligand descriptors) improves the operating point but consumes the five-sample ensemble and cross-model features, so it needs N model inferences per target. We report the native-only guarantee prominently and flag the combined score as the upgrade. The combined score's cross-model features are moderately correlated with novelty ($|\rho|$ up to 0.35), so it is a partly novelty-aware re-score; this is legitimate and places the *combined* score on the achievability side of the theorem below, which governs the frozen *native* score.

Gate. The gate accepts when $s \geq \tau$ and picks the largest accept set whose selective risk is certified $\leq \alpha$ by Learn-then-Test with fixed-sequence testing [1], testing each $H_0(\tau) : P(L{=}1 \mid s \geq \tau) \geq \alpha$

Table 2: Label provenance. The primary selective-risk results use the shipped BiSyRMSD label; only the label-free danger-floor / cross-model consensus analysis (SI) uses recomputed pose geometry, on the single-chain subset that App. C shows is the only frame that is trustworthy.

Result	Label / geometry	n
Break, frontier, repair, utility (Secs. 3–7)	shipped BiSyRMSD	11,254 target-labels
Cross-model danger floor / consensus (SI)	recomputed, single-chain receptors	6,163 instances
Interaction-fingerprint recovery (Sec. 7)	recomputed contacts vs. crystal	11,711 poses

with an exact binomial tail along a pre-specified, data-independent coverage grid (top-k ranks, $c \in \{0.05, 0.06, \dots, 1.00\}$, fixed before any risk was computed and archived in the repository). This gives $P(\text{selective risk} \leq \alpha) \geq 1-\delta$, finite-sample and distribution-free.

3 The exchangeability break

The break. A single global threshold hides severe per-stratum under-coverage. For AF3 the marginal realized risk $R_{\text{marg}} = 0.177$ masks $R_{\text{marg},S_3} = 0.375$ on the novel ligand stratum (Fig. 2, with Clopper-Pearson intervals and accepted n per point). The operational failure is starker: calibrating the gate on the familiar strata (S_0, S_1) and deploying on the novel tail (S_3, S_4) drives the realized error to $R_{\text{transfer}} = 0.55$ (AF3; 0.48–0.66 across the five models) against the 0.20 target, so the certificate is inverted, not merely loose. We lead the break on S_3; the no-analog S_4 (AF3 $n \approx 76$, greyed in the figure) is small-sample and enters only in the feasibility frontier, where zero coverage is the point.

Why it breaks: concept shift, not recency. Weighted conformal would fix this if the shift were pure covariate shift, where $P(L \mid s)$ is stable and only the marginal $P(s)$ moves. We test that by measuring *reliability drift* $D(\nu) = \sum_{\text{bins}} w\,[P(L{=}0 \mid s, S_0) - P(L{=}0 \mid s, S_\nu)]$, the confidence-conditional correctness gap between the familiar reference and stratum ν. Drift is large on structural novelty and its 90% bootstrap interval excludes zero for every model (Protenix reaches $+0.63$ on pocket novelty, Chai $+0.52$); the $+0.63$ is not a sparse-bin artifact, as every confidence quintile behind it holds ≥ 103 novel-pocket targets and the top-confidence bin still recovers to only 0.54 correct against 0.95 on the familiar stratum. Confidence is optimistic on novel structure, the signature of concept shift. A signed accept-region decomposition makes this quantitative: for AF3 on S_3 the target-risk gap of 0.328 splits into a concept term of 0.309 (90% CI $[0.27, 0.35]$) and a covariate term of 0.018, which Sec. 5 turns into a theorem.

The temporal axis is a benchmark artifact, not a null result. Earlier framings contrasted a large structural drift with a near-zero temporal drift. That contrast is real but must be stated precisely: every Runs N' Poses structure is deposited after the 2021-era models' training cutoff, so the temporal axis ranks recency *among already-out-of-training structures* rather than in- versus out-of-training, and a flat drift there is a property of the benchmark, not evidence of temporal robustness. The one model with a genuine within-benchmark cutoff boundary is Boltz-2 (2023-06-30); across it the reliability drift is $+0.005$ (90% CI $[-0.075, 0.079]$), and its structural S_3 break is unchanged whether keyed on the 2021- or the correctly-referenced 2023-training similarity (0.364 vs. 0.367 correct). Structural similarity is therefore the operative shift variable, and we do not claim a temporal-generalization result.

4 The feasibility frontier

The break is a fixed-coverage statement; it leaves open whether α is reachable by lowering coverage, so we measure that directly. We fix the deployed rule on the familiar stratum and, per target stratum, report the largest source coverage $c^\star_g = \max\{c : R_{Q,g}(\tau(c)) \leq \alpha\}$ at which it still holds α, one delivered pose per (system, model). The frontier decays with novelty and then vanishes: for AF3 on the ligand axis $c^\star$ runs $1.00, 0.85, 0.75, 0.20, 0.00$ across S_0 to S_4 at $\alpha = 0.20$, and $0.65, 0.40, 0.10, 0.00, 0.00$ at $\alpha = 0.10$. Requiring at least 20 accepted targets to call a stratum feasible, 21 of 50 model-by-stratum cells (two axes $\times$ five models $\times$ five strata) admit no operating point at any coverage at $\alpha = 0.20$, and 26 of 50 at $\alpha = 0.10$; pooling the two levels gives 47 zero-frontier cells, of which **35** survive an exact Clopper-Pearson lower bound on $R_{Q,g}$ at every measurable coverage. A zero frontier is strictly stronger

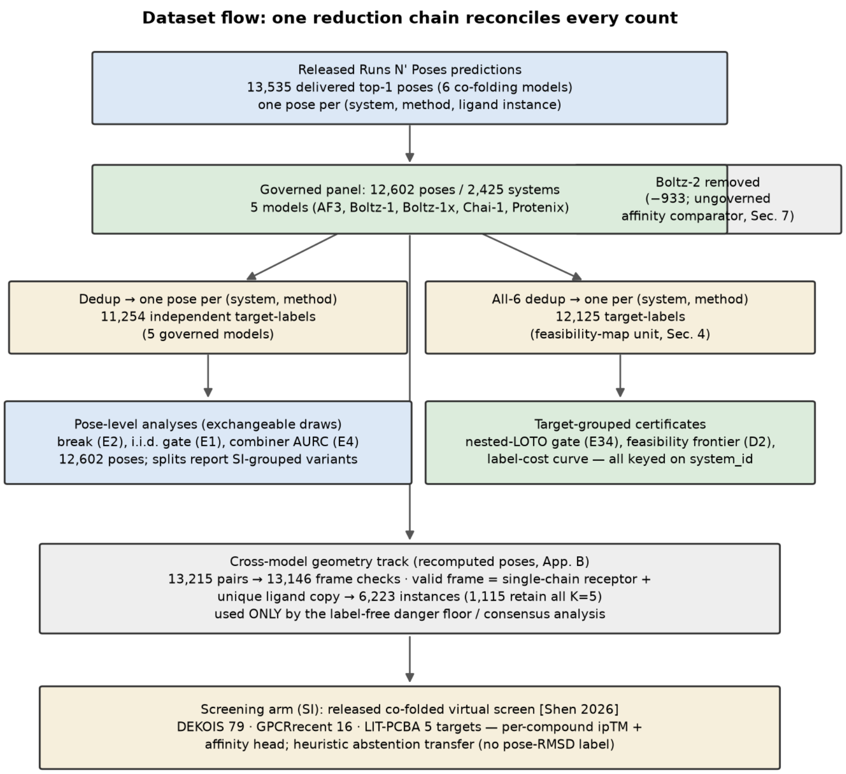

Figure 1: Dataset flow. One reduction chain reconciles every count; each node is annotated with the experiments that consume it. Counts are recomputed live from the released predictions.

than the coverage-pinned theorem of Sec. 5, which speaks only at the deployed coverage, and it is an empirical finding rather than a corollary. The infeasibility is not an artifact of the quartile binning: the zero-frontier fraction stays near 0.4 whether we use 3, 5, or 7 strata or fixed-similarity edges. The deployed view is starkest of all: the AF3 rule that Learn-then-Test certifies at $\alpha = 0.20$ on familiar targets realizes risk $R_{\text{gate},S_3} = 0.538$ on S_3, a margin of -0.338.

A one-sided escape survives the impossibility, which constrains only certified *upper* bounds: cross-model pose disagreement supplies a label-free *lower* bound on ensemble risk by triangle inequality to the crystal pose, extending an oracle-free disagreement floor known for discrete outputs [25, 26] and the folklore that pose agreement filters wrong poses [27] to a continuous structured target. It is blind to consensus error, where all models share one wrong mode: the consensus regime covers 64% of the analyzed complexes at risk 0.152 against 0.698 on the diverse remainder, and the hidden error grows from 0.058 on S_0 to 0.120 on S_3 even as consensus mass shrinks. The construction and its single-chain provenance are in the SI.

5 An exact decomposition, and what a training-free reweighting cannot fix

Fix the frozen score s and the accept rule $\{s \geq \tau\}$; write $Y = L$ for the error label. Let $\eta_P(x) = P(Y{=}1 \mid X{=}x)$ and η_Q be the error conditionals on the calibration law P and the deployment law Q, and call a reweighting *training-free* if its weights are covariate-measurable and use no target labels (density-ratio, classifier, or KLIEP weights, including the point mass of weighted conformal). Write $R_{\text{ref}}(\tau_c) = \mathbb{E}_Q[\eta_P \mid s \geq \tau_c]$ and $\bar{\Delta}_c = \mathbb{E}_Q[\eta_Q - \eta_P \mid s \geq \tau_c]$.

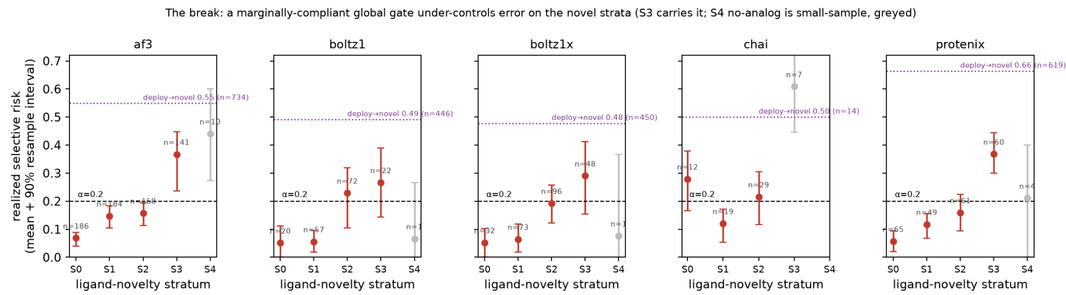

Figure 2: The break. Per-stratum realized selective risk of a marginally-compliant global gate, over 300 target-grouped calibrate/deploy resamples (mean, 90% resample interval, median accepted n per point); the no-analog S_4 is greyed as small-sample. Realized risk climbs above the target $\alpha = 0.20$ as ligand similarity falls, and the dotted line marks the calibrate-on-familiar / deploy-on-novel realized risk (AF3 0.55). The break is carried by S_3.

> **Proposition (exact decomposition; App. A).** At fixed coverage c the realized target selective risk is $R_Q(\tau_c) = R_{\text{ref}}(\tau_c) + \bar{\Delta}_c$, and $\bar{\Delta}_c$ is invariant to the choice of training-free weights. Hence the oracle-weighted plug-in certificate reports R_{ref} and under-states the realized risk by exactly $\bar{\Delta}_c$, and level α is unachievable at coverage c by any training-free reweighting whenever $\bar{\Delta}_c > \alpha - R_{\text{ref}}(\tau_c)$.

The content is not that α is achievable iff the risk is $\leq \alpha$ (definitional), nor that the accept set is fixed at fixed coverage (a one-line remark), but that the accept-region concept gap $\bar{\Delta}_c$ is *measurable and invariant*: the part of the error that comes from confidence meaning something different on novel molecules is exactly the part no unlabeled reweighting can see or remove. The proof is a tower-property identity: a covariate-measurable weight recovers only the source conditional η_P and cannot reach η_Q, completing the covariate-shift boundary of weighted conformal [3] and specializing the irreducible term of domain-adaptation lower bounds [23]; the same reweighting-invariant floor was shown concurrently for LLM abstention [19]. Matching achievability: in-stratum recalibration attains the floor R_{Q_g} given n_g stratum labels; pointwise conditional coverage being impossible [6], the guarantee is neighborhood-conditional, and its ratio functional is controlled with high probability by Learn-then-Test, whose finite-sample cost is exactly the label-cost curve of Sec. 6 rather than an asserted rate.

The decomposition's equalities assume the score has no atom at the operating threshold. On the primary `ranking_score` gate this holds exactly in the data (the ties mass at the deployed τ is 0 for every model); on the interface-ipTM feature the boundary-randomization bracket is at most 0.7% of coverage. So the identity is exact for the deployed native gate, with a $\leq 0.7\%$-coverage bracket noted where ipTM is used.

Validation. A synthetic generator with a known $P(Y \mid s, v)$ reproduces every term: $R_{\text{ref}} = 0.273$ and $\bar{\Delta} = 0.205$ sum to the realized $R_Q = 0.478$, the invariance of the realized risk across nine reweightings (residual < 0.001) is a regression test of the decomposition, and sweeping the target drift locates the coverage at which no reweighting reaches α. On the public non-co-folding benchmarks elec2 (worst-block concept gap 0.36) and ACS Income (gap 0.05), a single-threshold gate under-controls the worst block, weighted conformal issues a certificate its realized risk then violates on elec2 but repairs on ACS, and group-conditional calibration certifies control by abstaining (App. B). Details, including the elec2/ACS realized worst-block risks, are in the SI.

Drift selects the repair. Reliability drift $D(v)$ is a label-based diagnostic that tracks $\bar{\Delta}_c$, so it decides, from a labeled probe rather than after a failed deployment, which repair a region admits. A small drift signals a covariate-dominated shift that weighted conformal repairs by choosing a more conservative threshold from unlabeled data (weighted conformal's *only* lever is threshold selection; on the moderate ligand shift $S_0, S_1 \to S_2$ it pulls AF3 realized error from 0.27 at 98% coverage to 0.19 at 70% coverage). A large drift signals a concept-dominated shift where weighted conformal silently violates its own certificate, leaving group-conditional recalibration as the only valid repair.

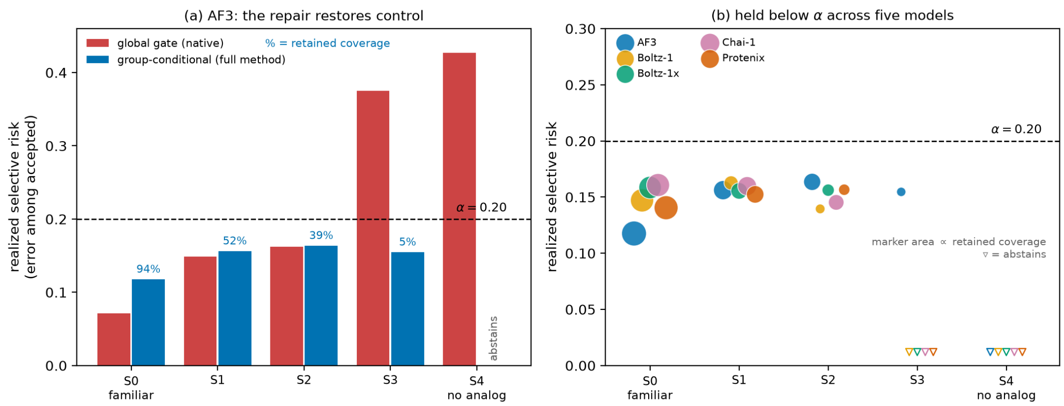

Figure 3: The repair. Group-conditional calibration holds realized selective risk at or below $\alpha = 0.20$ on the strata where the marginal gate over-shoots, folding to zero coverage on the no-analog tail rather than issuing a false certificate. Coverage labels are direct; the coverage cost grows with novelty.

Table 3: Baselines under the novelty shift (AF3, calibrate on familiar, deploy on novel; realized risk / coverage at $\alpha = 0.20$). The field baselines break; every gate that keys calibration on novelty holds by abstaining. Venn–Abers matches the combiner, so the shift repair does not require it.

	realized risk	coverage
Fixed ipTM ≥ 0.8 (field baseline)	0.301	0.69
Naive-transfer conformal	0.412	0.99
Accept-all	0.417	1.00
Group-conditional (combiner)	**0.190**	0.18
Localized / continuous novelty (RLCP) [5]	**0.187**	0.44
Per-stratum Venn–Abers [18]	**0.166**	0.24

6 Repairing the guarantee, and pricing it

Group-conditional calibration [17] keys a separate threshold per novelty stratum, restoring per-stratum control at an honest coverage cost (with the combined score AF3 accepts 52% of S_1 and 39% of S_2 at risk $\leq \alpha$, against 8% and 12% natively; Fig. 3). Where the marginal gate over-shoots α on the novel strata the per-stratum gate holds risk at or below it and, on the no-analog tail where no certified slice remains, folds to zero coverage rather than issue a false certificate. The repair does not require the gradient-boosted combiner: a per-stratum Venn–Abers recalibration of the native score reaches the same guarantee (AF3 realized risk 0.166 at 24% coverage vs. 0.190 at 18% for the group-conditional combiner), while the field baselines break under the same shift, a fixed ipTM ≥ 0.8 gate at realized risk 0.30 and naive-transfer conformal at 0.41 (Table 3).

The label cost, as a design curve. Group-conditional repair spends target labels, and how many depends on what is being certified. For the binary pose label the count is a sufficient statistic and the exact binomial tail is its exact inversion, so on the feasible strata it certifies at a median of 38 independent target-labels against 60 for a Hoeffding-Bentkus bound and 102 for pure Hoeffding. But 38 is a *familiar*-stratum price. Figure 4 turns the cost into a design tool: certified coverage as a function of the label budget n_g, per stratum. With the native score, no budget buys usable coverage on the novel strata, because there the rule fails, not the estimate of it, the impossibility made empirical. The combined score, a partly novelty-aware re-score and so on the achievability side, unlocks the moderate strata: AF3 reaches 22% certified coverage on S_1 at $\sim$80 in-stratum labels and 83% with the full pool, and 77% on S_2, while S_3 needs close to the full label pool for any coverage at all. The combined score raises the ceiling, not the label floor.

Deployment-computable stratifier. The repair needs a novelty stratum at deployment, but the oracle

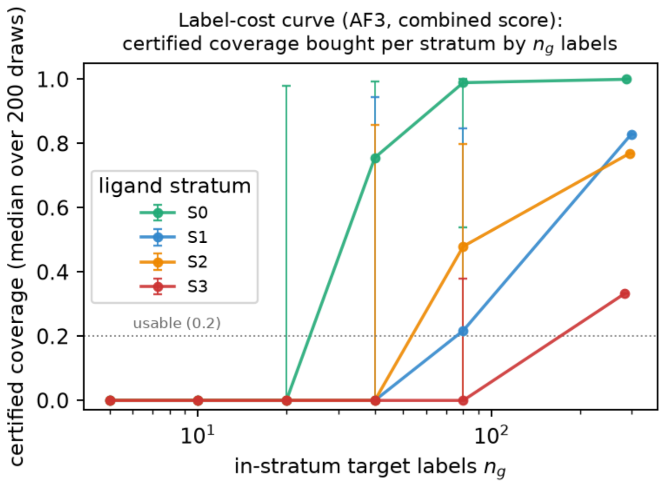

Figure 4: Label-cost curve. Certified coverage bought by n_g in-stratum target-labels, per ligand stratum (AF3, combined score, median over 200 draws with a 90% band). Moderate novelty (S_1, S_2) is certifiable at a measured label budget; the near-novel S_3 is label-starved even at the full pool.

similarity is computed against the model's training set, which for a closed model like AF3 a user cannot enumerate. A public proxy built from the count of pre-cutoff PDB analogs (deposition recency and protein sequence identity are at chance) recovers the oracle stratification with rank correlation 0.59 and 83% adjacent agreement, and matches oracle per-stratum control on the low and moderate strata, but loosens control on the near-novel tail (true-S_3 risk $0.23 \to 0.37$). The binding limit, however, is not the proxy: even the *oracle* group-conditional gate leaves the no-analog stratum above α (risk 0.60). So for closed-training models the repair is deployable one stratum out with a public proxy, and the extreme novel tail is uncertifiable by stratification alone, whether the stratifier is proxy or oracle. That, not a missing training set, is the honest ceiling.

A label-free worst-subpopulation certificate. When the novel subpopulation is unnamed we control the conditional value-at-risk of the accepted loss, the mean error of its worst $(1-\beta)$-fraction. For the binary label $\mathrm{CVaR}_\beta = \min(1, r/(1-\beta))$ with accepted error rate r, so a finite-sample bound on r [2] yields a CVaR_β bound at every β, and by CVaR–DRO duality this certifies that every accepted subpopulation of mass at least $m^\star = 1-\beta^\star$ has error $\leq \alpha$ [4], with no stratum labels and no importance weights, reparameterizing to a χ^2 robustness radius $\rho^\star$ [22]. A Bonferroni δ/K correction gives one certificate jointly across the five models (at 20% coverage, $m^\star \leq 0.52$ and $\rho^\star \geq 0.10$). This certificate is a real capability on moderate shift and, we report plainly, *vacuous exactly where it is most needed*: on deploy-to-novel $m^\star \to 1$ for most models, so the label-free certificate cannot rescue the novel tail either.

Match the certifier to the loss. On the *graded* loss $\min(\mathrm{RMSD}, 4)/4$ a variance-adaptive betting bound [28] certifies a usable operating point where a distribution-free bound certifies none: AF3 certifies 43% coverage at a 1 Å mean-RMSD target with realized capped-mean RMSD 0.90 Å, against essentially zero (0.04%) for a worst-case Hoeffding bound; the empirical guarantee coverage over 120 resamples is 0.90. The reliability layer is therefore not tied to the 2 Å convention, and "1 Å" is an actionable operating point where "$\alpha = 0.20$" is not.

Certificate ledger. Table 4 states, per claim, the δ, the multiplicity method, the n, and whether the claim is joint or per-model, so no favorable δ is chosen per claim. The 55-cell drift grid (5 models $\times$ 11 axis-strata: 4 ligand + 4 pocket + 3 temporal non-reference strata) uses a Romano-Wolf step-down bootstrap [29] at family-wise 0.10; all 40 structural cells survive, an effect-dominated family, and only 2 of 15 temporal cells do.

Table 4: Certificate ledger. $K = 5$ models; joint claims split δ by a Bonferroni union bound.

Claim	δ	multiplicity	n (unit)	scope
Global / LOTO gate	0.10	fixed-sequence LTT	target-labels	per-model
Joint LOTO gate	0.02	Bonferroni δ/K	target-labels	joint (5 models)
CVaR / DRO bound	0.02	Bonferroni δ/K	accepted poses	joint (5 models)
Betting bound (graded)	0.10	anytime-valid (Ville)	accepted poses	per-model
Drift grid	0.10	Romano-Wolf step-down	55 cells	family-wise
"For every model" effects	0.10	intersection-union	per-model p	no penalty

Table 5: Leakage-free nested-LOTO gate, native vs. combined score ($\alpha = 0.20$, $\delta = 0.10$): coverage, realized risk, exact Clopper-Pearson upper bound, and folds holding. Chai/Protenix native gates abstain.

	native			combined		
	cov	risk	$\mathrm{CP}^{\uparrow}_{90}$	cov	risk	$\mathrm{CP}^{\uparrow}_{90}$
AF3	0.20	0.189	0.218	0.73	0.176	0.190
Boltz-1	0.21	0.188	0.219	0.60	0.181	0.199
Boltz-1x	0.20	0.164	0.194	0.48	0.182	0.202
Chai-1		abstains		0.64	0.178	0.195
Protenix		abstains		0.57	0.182	0.199

7 Utility, priced and leakage-free

Ranking utility. Every utility number is target-grouped. Under a nested leave-one-target-out protocol (outer GroupKFold on system id; within each training fold a grouped fit/calibration split, so the combiner, the threshold, and the test target are mutually disjoint), the combined-score gate keeps **73**% of AF3 poses at certified risk $\leq \alpha$ (realized 0.176, Hoeffding-Bentkus upper bound 0.192), where the native gate keeps **20**%; at the stringent $\alpha = 0.10$ the combined gate keeps 13% at realized risk 0.066. Chai and Protenix native gates abstain entirely under this leakage-free protocol, the honest native limit the combined score recovers (64% and 57% coverage). These replace an earlier held-out-split pair (71% vs. 22%) that a pooled split would have leaked; the grouped protocol shares zero targets between calibration and test, where a random row split leaks 95%. The leave-one-target-out risks come with certified upper bounds and pass/fail per model (Table 5).

A decision curve retires the choice of α. Rather than defend a level, we sweep the cost ratio $\lambda = \mathrm{cost(act\ on\ a\ wrong\ pose)}/\mathrm{cost(abstain\ on\ a\ right\ pose)}$ and plot net benefit (Fig. 5). The conformal gate is best for $\lambda \in [0.65, 4.55]$ (AF3; every model shows the same window), dominating a fixed-ipTM threshold at *every* λ and beating accept-all once wrong poses are more than $\sim 0.65\times$ as costly as a missed right one. The layer's job is the high-cost regime, which is where a delivered pose that must be trusted lives.

A non-circular quality, honestly conditioned. Accepted-pose purity is $1-$ selective risk by construction, so it cannot show the gate buys anything past the guarantee. Interaction-fingerprint recovery, the fraction of the crystal's protein-ligand contacts a pose reproduces, is a different quantity a medicinal chemist reads. Pooling all accepted against all rejected poses shows a large gap (0.19 for AF3), but most of that is the accepted set being enriched for low-RMSD poses. Conditioning removes the confound: *within* the sub-2 Å set the accepted-minus-rejected recovery gap is $+0.02$ to $+0.04$, and a regression of recovery on RMSD with the gate as a covariate leaves a gate coefficient of $+0.03$ to $+0.07$ (all five 90% intervals excluding zero). So the gate buys a small but robust non-circular interaction-recovery lift, once the RMSD confound is removed. Separately, the accepted set is PoseBusters-valid at a higher rate than the rejected set for every model (AF3 0.77 vs. 0.58), a free strengthening; certifying the joint label (RMSD ≤ 2 Å *and* physically valid) directly is expensive because the confidence ranks for correctness, not validity, and we report that cost rather than hide it.

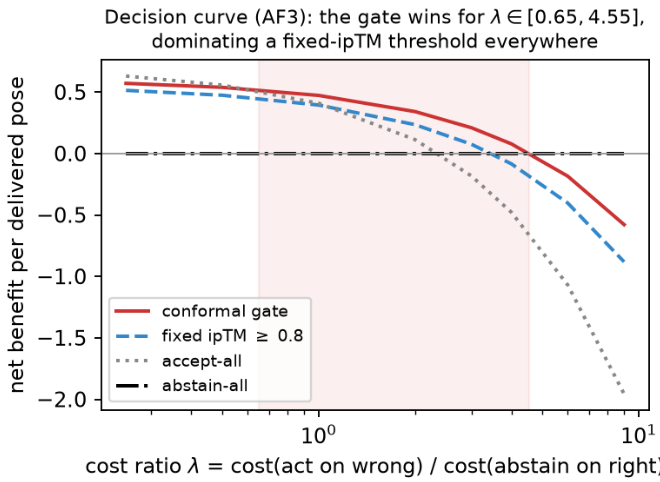

Figure 5: Decision curve (AF3). Net benefit per delivered pose vs. the cost ratio λ; the shaded band is where the conformal gate is the best of the four strategies. It dominates a fixed-ipTM threshold everywhere and serves the high-cost regime, so there is no single α to defend.

8 Generality and limitations

Pseudo-prospective. Freezing calibration on everything deposited before 2023-05-31 and deploying on everything after, with no stratum information at calibration, the native gate holds realized post-cutoff risk $\leq \alpha$ (or its Clopper-Pearson upper bound does) for four of five models and the combined gate for three, with realized-minus-expected risk in $[-0.04, +0.00]$; the abstentions are honest non-certifications, not violations. This is a recency transfer, complementary to the structural story, and inherits the caveat that Runs N' Poses is entirely post-cutoff.

Cross-dataset. Recast as risk control rather than ranking, a threshold frozen on Runs N' Poses and deployed on regenerated FoldBench [9] Protenix predictions does not overshoot α on either subset, but it certifies almost nothing on the 52 low-homology targets (accepts 3, coverage 6%, Clopper-Pearson risk upper bound 0.63), so the failure mode of cross-dataset transfer is coverage starvation, not risk violation, the coverage collapse of Sec. 3 reproduced on a second dataset. The interface-ipTM gate out-ranks a `ranking_score` control (AURC 0.380 vs. 0.454) but their 90% bootstrap intervals overlap, and the regenerated model's self-scored top-1 success (0.40) differs from the released 0.57, so this stays a directional transfer check, feature and label self-consistent within the one run. A third dataset would need a benchmark that releases ipTM and PoseBusters per pose; where none is available we state that concrete community ask rather than overclaim generality.

Limitations. The no-analog tail is uncertifiable by every method, where abstention is the only action. The best ranker in our screening analysis (SI), a Boltz-2 affinity head, carries no coverage guarantee: extending the framework to certify an affinity ranker (which needs an affinity-error label) is the top open problem. All evaluation is retrospective; a prospective screen with wet-lab confirmation is the natural next step.

9 Conclusion

Co-folding confidence becomes a risk-controlled decision only when the guarantee is audited where it fails. On structural novelty a marginally valid gate under-controls error more than twofold, and on 35 of 47 zero-frontier cells no threshold on the frozen score is certifiable at any coverage. An exact decomposition explains why: the concept gap that opens there closes under in-stratum recalibration but under no training-free reweighting. Three honest bottom lines follow. Co-folding confidence is certifiable on the familiar strata. It is extendable one stratum outward at a measured price, a median of 38 in-stratum target structures, and with a public proxy stratifier for closed-training models. And on the extreme novel tail, exactly where a program most needs a guarantee, neither stratification nor a label-free certificate can supply one, so abstention is the certified action. The layer runs on frozen models' released predictions on CPU and ships as certificate cards per model and novelty stratum.

Data and code availability. Code and calibration tables are at `https://github.com/rikhinkavuru/foldgate` under an OSI-approved license; a one-command `make repro` reproduces the tabular results from a pinned environment (`uv.lock`, fixed seeds), with the screening, cross-model, and structure-feature arms opt-in because they consume larger released artifacts. Runs N' Poses is at Zenodo (`10.5281/zenodo.14794785`); the regenerated FoldBench predictions and self-scored labels are at Zenodo (`10.5281/zenodo.21417241`); the screening scores are from **(author?)** [10]. The coverage grid, fixed-sequence order, binning, and endpoints are frozen in a timestamped pre-registration file in the repository.

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A Proof of the decomposition, and the ties bracket

Setup. Frozen score $s = s(X)$, error label $Y \in \{0,1\}$, accept region $A_\tau = \{s \geq \tau\} \in \sigma(X)$. A training-free reweighting has weights $w = w(X) \geq 0$ that are $\sigma(X)$-measurable and use no target labels. Assume s has no atom at the operating threshold, so for each coverage c there is a unique τ_c with $Q(s \geq \tau_c) = c$; real scores carry ties handled by boundary randomization, under which the equalities become $\leq$ / $\geq$ brackets, and Sec. 5 bounds the bracket empirically (0 on the ranking-score gate, $\leq 0.7\%$ coverage on ipTM).

Lemma (reweighting sees only the source labels). For any such w, the reweighted plug-in selective risk is $\widetilde{E}^w(\tau) = \mathbb{E}_P[wY\mathbf{1}_{A_\tau}]/\mathbb{E}_P[w\mathbf{1}_{A_\tau}] = \mathbb{E}_{Q_w}[\eta_P \mid A_\tau]$, with $dQ_w \propto w\,dP_X$. *Proof.* w and $\mathbf{1}_{A_\tau}$ are $\sigma(X)$-measurable, so tower over X: $\mathbb{E}_P[wY\mathbf{1}_{A_\tau}] = \mathbb{E}_P[w\eta_P\mathbf{1}_{A_\tau}]$; normalize. The target conditional η_Q cannot appear. □

Proposition. At coverage c: (a) $R_Q(\tau_c) = R_{\text{ref}}(\tau_c) + \bar{\Delta}_c$ with $R_{\text{ref}}(\tau_c) = \mathbb{E}_Q[\eta_P \mid A_{\tau_c}]$ and $\bar{\Delta}_c = \mathbb{E}_Q[\eta_Q - \eta_P \mid A_{\tau_c}]$; (b) any w realizing coverage c accepts exactly A_{τ_c}, so its realized risk equals $R_Q(\tau_c)$, independent of w; (c) with oracle weights the certificate of the Lemma equals $R_{\text{ref}}(\tau_c)$, under-reporting the realized risk by exactly $\bar{\Delta}_c$; (d) level α is achievable at coverage c by some training-free reweighting iff $\bar{\Delta}_c \leq \alpha - R_{\text{ref}}(\tau_c)$. *Proof.* (a) $A_{\tau_c} \in \sigma(X)$, so $R_Q(\tau_c) = \mathbb{E}_Q[\eta_Q \mid A_{\tau_c}]$; add and subtract $\mathbb{E}_Q[\eta_P \mid A_{\tau_c}]$. (b) The no-atom assumption pins τ_c from c; realized risk is a functional of (Q, τ_c) alone. (c) Lemma with $w^\star = dQ_X/dP_X$ gives $Q_{w^\star} = Q$. (d) is (a)+(b). □ The statement is coverage-pinned and binds only scalar thresholds of a fixed score with covariate-measurable weights; re-scoring on ν or a per-stratum threshold escapes it (the achievability side), and in-stratum split calibration [21] gives stratum-conditional miscoverage $\leq 1/(n_g+1)$ regardless of drift, with the ratio functional controlled by Learn-then-Test at the finite-sample cost the label-cost curve measures.

B Benchmark and validation details

The synthetic generator draws a stratum from a tilted ν-marginal, a score logit $z \sim \mathcal{N}(m_0 - E\nu, \sigma^2)$, and a label with $P(Y{=}1) = \varepsilon + (1-2\varepsilon)\,\text{sigmoid}(\beta_0 + \beta_1 z - D_{\text{eff}}\nu)$, isolating covariate shift (E), concept drift (D), and an error floor (ε); every proposition quantity is available in closed form and cross-checked by Monte Carlo. The real benchmarks reduce any classifier to (s, y, ν); realized worst-stratum selective risk at $\alpha = 0.20$: elec2 marginal 0.31, weighted-conformal 0.34, group-conditional 0.20; ACS Income marginal 0.23, weighted-conformal 0.20, group-conditional 0.20, the reliability drift separating the two being 0.36 (elec2, concept) against 0.05 (ACS, covariate).

C A benchmark-integrity note for cross-model pose geometry

Recomputing pose RMSD in a common frame carries a silent trap. Runs N' Poses ships only the system's receptor chain while co-folding models predict the full biological assembly, so a chain-ordinal superposition can land the ligand in the wrong protomer of a homodimer: the backbone still fits (superposition residual ~ 0.3 Å) while the ligand sits tens of Ångström away. Neither the residual nor a triangle-inequality consistency check detects it, and a distance-consistency check passed at $0/13{,}146$ while 21% of recomputed distances and 15% of binary correctness labels were wrong. Only comparison against the shipped BiSyRMSD label exposes it; restricting to single-chain receptors with an unambiguous ligand copy raises agreement from Spearman 0.620 to 0.995 and defines the 6,163-instance subset any recomputed-geometry result in this paper uses (Table 2). We flag this for reuse; a symmetry-aware chain assignment is the general fix.